Nhat-Quynh **Le-Pham**

PhD student in Computer Science at Trinity College Dublin

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Summary

My research interests focus on artificial intelligence, computer vision and computer graphics, with a particular emphasis on **3D understanding** and **3D/4D generative modeling**. I am interested in building models that can perceive, represent, and generate dynamic 3D worlds.

Education

Trinity College Dublin — *Dublin, Ireland*

Doctor of Philosophy in Computer Science — *Sep 2026 - Sep 2030*

- Research area: Agentic 3D Reconstruction

University of Science - Viet Nam National University Ho Chi Minh City — *Ho Chi Minh, Vietnam*

Bachelor of Science - Advanced Program in Computer Science (APCS) — *Oct 2020 - Dec 2024*

- **GPA: 3.8/4.0, Thesis: 10/10** | Language of instruction: English
- Awarded **Top GPA Scholarship** (2023)

Experiences

Qualcomm AI Research — *Ho Chi Minh, Vietnam*

Research Resident | Mentor: Prof. Binh-Son Hua, Dr. Khoi Nguyen — *Feb 2025 - Feb 2026*

- Conducted research on open-vocabulary 3D understanding and world models with leading research scientists and professors.
- Developed and implemented efficient 3D perception and generative models optimized for edge and mobile devices.

VinAI Research — *Ho Chi Minh, Vietnam*

Research Resident | Mentor: Prof. Binh-Son Hua, Dr. Trung-Nghia Le — *Aug 2023 - Feb 2025*

- Conducted independent research for the graduation thesis under the guidance of faculty members from my home institution and professors from Trinity College Dublin.
- Proposed a novel text-to-4D content generation using only image priors and monocular driver videos, bypassing the heavy computational cost of video diffusion models.

Software Engineering Lab (SELab) - University of Science - VNUHCM — *Ho Chi Minh, Vietnam*

Research Student | Mentor: Mr. Hai-Dang Nguyen — *Jun 2021 - Mar 2023*

- Led and coordinated a student research group participating in international AI challenges and workshops in object detection, OCR, and AI for smart city applications.
- Conducted research on fake news classification and 3D shape retrieval, resulting in multiple peer-reviewed workshop and journal publications, including MediaEval 2021 and Computers & Graphics at 3DOR 2022 and 3DOR 2023.

MTI Technology — *Ho Chi Minh, Vietnam*

Data Scientist Intern — *Jun 2021 - Dec 2021*

- Collaborated with a five-person team to develop a Japanese drug bill information retrieval system via optimizing Detectron2 and Mecab.
- Improved the accuracy under ambient lighting and shaking-quality noise conditions via heuristic approach.

Publications

TruncGradGS: Improved 3D Gaussian Splatting via Truncated Gradient Updates — *Pacific Graphics 2026*

Theo Morales, **Nhat-Quynh Le-Pham**, Robin Atkins, Binh-Son Hua — *Oct 2026*

- Proposed a piecewise truncated gradient formulation to mitigate gradient vanishing in 3D Gaussian Splatting, improving optimization stability and robustness to initialization across static and dynamic scenes.
- Introduced a **synthetic dynamic Gaussian Splatting benchmark dataset and evaluated the method on public and proposed benchmarks.

Prepare Lighter and Faster for Open-Vocabulary Queries: A Query-Wise 3D Segmenter for Gaussian Splatting — *Under Review*

Nhat-Quynh Le-Pham, Khoi Nguyen, Binh-Son Hua — *May 2026*

- Proposed a novel on-device, lightning-fast query-wise open-vocabulary 3D instance segmentation on 2D/3D Gaussian Splattings, achieving state-of-the-art performance in both runtime efficiency and visual quality.

Text-to-4D Content Creation using Image Priors and Monocular Driver Videos *Bachelor Thesis*

Nhat-Quynh Le-Pham - Thesis | Video *July 2024*

- Developed an end-to-end framework for controllable, high-quality 4D object generation from text prompts using only image priors and driver videos, removing reliance on generative video models.
- Introduced efficient image prior fine-tuning, score distillation, and attention swapping to transfer motion with high fidelity.
- Defended the graduation thesis with **distinction**, earning a perfect score of **10/10 (Top 1%)**.

Patents

Query-based Open-vocabulary 3D object segmentation from Gaussian Splats *US Patent*

Nhat-Quynh Le-Pham, Khoi Nguyen, Binh-Son Hua *Nov 2025*

Workshops & Journals

SketchANIMAR: Sketch-Based 3D Animal Fine-Grained Retrieval *3DOR'23*

Published in Computers & Graphics Journal, Elsevier (Q1) - Paper | Github *Mar 2023*

- Developed a cross-domain contrastive learning framework aligning 3D objects with text via cosine similarity, using an EfficientNetV2-Small–based feature extraction pipeline with Canny edge and KMeans-based dataset augmentation, improving Nearest Neighbor performance by 0.5.

TextANIMAR: Text-Based 3D Animal Fine-Grained Retrieval *3DOR'23*

Published in Computers & Graphics Journal, Elsevier (Q1) - Paper | Github *Mar 2023*

- Developed a cross-domain contrastive learning framework aligning 3D objects with text using cosine similarity, with a CLIP and EfficientNetV2-Small–based feature extraction pipeline and KMeans-based dataset augmentation, improving Nearest Neighbor performance by 0.5.

SHREC'22: Fitting and recognition of simple geometric primitives on point clouds *3DOR'22*

Published in Computers & Graphics Journal, Elsevier (Q1) - Paper | Github *Feb 2022*

- Built recognition method of geometric shapes on 8100 point clouds based on PointNet to extract global and local features for classification, followed by primitive-specific shape parameter fitting using least squares and geometric methods, with final predictions determined via majority voting among the top models.

FakeNews: Corona Virus and Conspiracies Multimedia Analysis Task *MediaEval'21*

Published in MediaEval 2021 Workshop - Paper *Dec 2021*

- Developed a COVID-19 conspiracy misinformation framework using a multimodal ensemble of BERT, Naive Bayes, and LSTM, combined with extensive tweet preprocessing and dataset augmentation, achieving promising results despite challenges from dataset imbalance and sentiment bias.

Scholarships & Awards

2023 **Vietnam Female Students in Science and Technology Award,** Ministry of Science and Technology *Ha Noi, Vietnam*
Top 20 female sci-tech students of 2023

2023 **Top GPA Scholarship,** Faculty of Information Technology - University of Science - VNUHCM *HCMC, Vietnam*
Top 1%

2023 **First Prize,** SHREC 2023: Sketch-Based 3D Animal Fine-Grained Retrieval *3DOR'23*

2023 **First Prize,** SHREC 2023: Text-Based 3D Animal Fine-Grained Retrieval *3DOR'23*

2022 **Ranked 3rd,** SHREC 2022: Fitting and recognition of simple geometric primitives on point clouds *3DOR'22*

2021 **Finalist,** Ho Chi Minh City AI Challenge *HCMC, Vietnam*

Certificates

IELTS 7.0

References

Prof. Binh-Son Hua Assistant Professor at Trinity College Dublin - Homepage | Google Scholar | Email

Dr. Khoi Nguyen Staff Research Scientist at Qualcomm AI Research - Homepage | Google Scholar | Email

Dr. Trung-Nghia Le Senior Researcher & Lecturer at University of Science - VNUHCM - Homepage | Google Scholar | Email

Mr. Hai-Dang Nguyen Researcher & Lecturer at Software Engineering Lab (SELab) - University of Science - VNUHCM - Google Scholar | Email